A child lives with the young

The rule, the two periods written out, and what full weight on the child costs

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1 The rule

> *"maybe the right way to do this is to have the parent to have a child only when they are young and have no kids when they are old. then children are only with the young! and the young transmit the SUD penalty only if they get it in the transition to being old. The old dont weight the child at all since it left, so they only care about their own consumption and not the child's."*
>
> *"if the old die with SUD, that does not transmit to the child, since the child is already young at that time."*

A child is in the household for exactly one period, the parent's period 1. It leaves at the end of it, and it leaves as a young adult, which is why the cohort arriving each period has mass one. Everything that happens to the parent after that happens to somebody the child no longer lives with. So the equivalence scale is $\xi = \sqrt{2}$ in period 1 and 1 in period 2; the old carry no child term; and the last moment that can transmit anything to the child is the *transition* out of period 1, where the disorder is realised.

The demography closes exactly. At a date there is mass 1 of children, mass 1 of young, and mass $\bar{s} = \phi_0(1 - \kappa\bar{\pi})$ of old. This period's children are next period's young, so the cohort entering is mass 1 forever: no child is ever between households, and no period of a life is spent in nobody's household. That is what fails if a child straddles the transition — then a parent's death truncates a childhood and the childhood flow has to be survival-weighted.

The weights add up, which is why the child's weight is $\sqrt{2}-1$. A household of two counts as $\xi = \sqrt{2}$ adult equivalents and reaches the utility of one person holding $1/\xi$ of its resources. Give the parent weight 1 and the child $\xi - 1$ and the weights sum to ξ exactly: with no drugs $X_P = X_K = X$ and $u(X_P) + (\xi - 1)u(X_K) = \xi\, u(c/\xi, h/\xi)$, so nothing is created or lost by the household having a child. A weight of 1 would count two people in full while deflating their bundle as though there were $\sqrt{2}$.

2 The two periods, written out

Period 2 — terminal, no child. State: assets a, income y, disorder $d' \in \{0, 1\}$. Employment is drawn (u_0 clean, u_1 with the disorder), a voucher is drawn subject to the policy rule, housing is

chosen from {large rental, small rental, street}. Assets are run down and nothing is bequeathed, so $c = Ra + \text{inc} - \text{rent}$. There is no drug choice.

$$X_{P2} = c^{\varphi_c} h^{\varphi_h} \bar{b}^{\,\varphi_q}, \qquad W_2(a, y, d') = \max_h \; u(X_{P2}) + \aleph, \qquad u(x) = \tfrac{x^{1-\sigma}}{1-\sigma}.$$

$\xi = 1$ and the child term is absent. This is already the rule.

Period 1 — the child is present. State: income y, an employment draw and a voucher draw, giving $(\text{inc}, r_{\text{mul}})$. The household chooses $v \geq 0$, housing and saving a', with $c = \text{inc} - p_v v - \text{rent} - a'$ and $\xi = \sqrt{2}$:

$$X_P = \left(\tfrac{c}{\xi}\right)^{\varphi_c} \left(\tfrac{h}{\xi}\right)^{\varphi_h} (\bar{b} + v)^{\varphi_q}, \qquad X_K = \left(\tfrac{c}{\xi}\right)^{\varphi_c} \left(\tfrac{h}{\xi}\right)^{\varphi_h} \bar{b}^{\,\varphi_q}.$$

The child's bundle carries the Stone–Geary shifter alone: it consumes none of the drug, so the drug reaches it only by taking consumption out of the shared c. Onset at the transition is $\pi(v) = 1 - e^{-\lambda v}$, and survival is $s_0 = \phi_0$ clean, $s_1 = \phi_0(1 - \kappa)$ with the disorder, *inside* the expectation. The household maximises

$$V \;=\; \underbrace{u(X_P) + \omega_c(\xi - 1)u(X_K) + \aleph}_{\text{flow}} + \beta\big[(1-\pi(v))s_0 W_2(a', y, 0) + \pi(v) s_1 W_2(a', y, 1)\big] - \underbrace{\omega_c \cdot \texttt{chdmg} \cdot \pi(v)}_{\text{continuation half}}.$$

So ω_c acts in **two** places, and only one of them is switched on. `chdmg` is the child's expected lifetime loss, it is declared 0.0 at `stylized.py:641` and nothing sets it. The continuation half has never been live in any calibration; the reduced form was a placeholder for a child value function that was never built.

3 Full weight on the child

> *"the young household should put the full weight on their child both in contemporaneous consumption and the future value of their life because they become adults right after."*

That is $\omega_c = 1$ in the flow, and `chdmg` replaced by the child's actual value function. The child becomes an adult exactly when the parent becomes old, so both arrive one period later and take the same β:

$$\boxed{V_1(y) = u(X_P) + (\xi - 1)u(X_K) + \aleph \;+\; \beta\, \mathbb{E}_{d'}\big[\phi(d') W_2(a', y, d')\big] \;+\; \beta\, \mathbb{E}_{d'}\big[V_1(y'(y, d'))\big]}$$

$$\log y' = \rho \log y + (1 - \rho) \log \bar{y} - \delta_d \mathbf{1}\{d' = 1\}, \qquad \rho = 0.35.$$

The child's survival is not the parent's — the child lives whether or not the parent dies — so $V_1(y')$ carries no $\phi(d')$; only W_2 does. And `chdmg` falls out as what it always meant, $V_1(y'_{\text{clean}}) - V_1(y'_{\text{pen}})$.

Four things it forces.

1. **It becomes a fixed point.** V_1 is on both sides, so period 1 is no longer a backward solve: it needs iteration to a fixed point over the income grid. At $\beta = 0.86038$ the dynastic multiplier is $1/(1 - \beta) = 7.16$, so the level of V_1 rises by roughly seven-fold and the CEV, which needs a level, has to be recomputed against it. The mortality trade-off is unaffected: the dynastic term does not contain ϕ_0, so it drops out of the VSL derivative.

2. ω_c **is not a free dial.** It is calibrated, identified on the child use gap, and came out at 0.01518. Setting it to 1 overrides a moment the model currently matches to +0.3 per cent.

3. **Full internalisation removes the externality.** At $\omega_c \approx 0$ the damage to the child is a pure externality and there is a child-based case for policy. At $\omega_c = 1$ the parent already weighs the child's loss in full when choosing v. That may be the benchmark you want, but it is the mechanism, not a detail.

4. **Welfare then double-counts.** If the parent internalises the child fully, V_1 already contains the child's whole lifetime, so summing V_1 over cohorts counts each generation repeatedly. The code half-handles this and handles it imperfectly: `own = tot + pen` adds the continuation penalty back but leaves $\omega_c(\xi - 1)u(X_K)$ inside V_1, so childhood is counted $1 + \omega_c$ times — a 0.5 per cent overcount today, a full double count at $\omega_c = 1$.

The argument in its favour. $\omega_c = 1$ is the only value consistent with the equivalence scale. The household's weights then sum to $1 + (\xi - 1) = \xi = \sqrt{2}$, the same $\sqrt{2}$ that deflates its bundle. Any $\omega_c < 1$ means the household spends as $\sqrt{2}$ people and is counted as fewer.

4 $\beta = 0.995$ annual, and what it costs before recalibration

$0.995^{30} = 0.86038$ over a thirty-year period. That is what `stylized.py` carries as its default and what the September 13 sweep overrode to $\beta = 1.0$, at its upper bound. Setting it back is necessary for the dynastic term — at $\beta = 1$ the sum $V_1 = \text{flow} + V_1(\text{child}) + V_1(\text{grandchild}) + \ldots$ adds about 9.6 per generation forever and does not converge.

It cannot simply be substituted in. Holding every other parameter at the calibrated vector and moving β alone:

moment	target	$\beta = 1.0$	$\beta = 0.86038$	gap at 0.86038
drug participation	0.08901	0.08871	0.32387	+264%
population SUD rate	0.03680	0.03668	0.13392	+264%
users' income / average, r_{use}	0.74000	0.73526	**1.77923**	+140%
drug spending / income, users	0.14098	0.14132	0.15855	+12.5%
value of a statistical life	80.000	79.999	68.829	−14.0%
SUD share of the homeless	0.21700	0.21701	0.24871	+14.6%
housing / non-drug spending	0.26800	0.26686	0.28077	+4.8%
homelessness rate	0.01000	0.00962	0.01026	+2.6%

The direction is the model's own logic — a less patient household weighs a thirty-year consequence less and uses more — but the size of it matters, and $r_{\text{use}} = 1.78$ is the tell. **Users richer than average is the corner branch**, the one `recalibrate.py` records a failed sweep on: *"participation 66 per cent, users RICHER than average at r_{use} 1.36, with four of the eight parameters cornered on their bounds trying to climb back."* So $\beta = 0.86038$ needs a full recalibration, and it must not be started from the $\beta = 1$ vector, because the same file records that the starting point is what picks the branch. Its interior-branch start is $h_{hl} = 10^{-4}$, $\aleph = 15.0$, $\theta = 0.16$, $u_1 = 0.127$, $\pi_{\text{base}} = 0.40$, $\bar{b} = 0.0282$.

The childhood level barely moves (1.43349 → 1.41912), because a childhood is period-1 flow and carries almost no discounting. The adult value falls 8.19940 → 6.77100, which is the continuation being discounted.

5 Three of the four parts are already built; one is contradicted

what the rule requires	where it lives	state
$\xi = \sqrt{2}$ in period 1, 1 in period 2	`stylized.py:882 _flow`	built
the old carry no child term	`stylized.py:972`, `child=False`	built
the SUD transmits on the transition out of period 1	`transition.py:119`, `use×prog`	built
nothing after the child leaves transmits	`transition.py:119`, `ev`	**contradicted**

The child's penalty has two legs and they key on different periods. The **SUD leg** is `use_prob×prog`: the chance the parent uses in period 1 times the chance a user acquires the disorder at the transition. That is the rule already. The **homelessness leg** is `terminal_homeless_prob`, the parent's chance of being on the street *in period 2* — under the rule, exactly the state that cannot reach a child who has left.

Period-2 keying was deliberate: it was put there so Treatment First's own revocations could reach a child, since the voucher is withdrawn at the transition and period-1 homelessness cannot register it. `TRANSITION_LESSONS.md` carries it as a standing instruction. Both cannot hold, so **nothing has been changed**. Note that keying on period 1 does not switch conditionality off: it still reaches the child through the SUD leg by moving how much people use. What is lost is the homelessness half of the revocation.

6 "SUD transmission is smaller" — where that bites

Not in the stylized model. Drugs are chosen in period 1 only and the disorder is realised at the transition, so there is no old-acquired SUD to remove: the transmitted object already *is* the period-1 one, and the share of children born behind the SUD penalty stays at 3.86 per cent.

It bites in the *quantitative* model, where use is chosen at every age and a parent can acquire the disorder long after its children have left. There the rule genuinely cuts the transmitting population. Expect δ_d to rise when it does: the penalty is calibrated to the child's SUD odds ratio, so a smaller transmitting share is met by a larger fitted penalty and the product stays pinned by the target. This has not been measured.

7 The numbers, at the calibrated $\beta = 1$ vector

the child in the household		**what the child inherits**	
equivalence scale ξ	1.41421	participation	0.08871
child's weight $\xi - 1$	0.41421	of users, acquire the disorder	0.41348
parent's weight	1.00000	population SUD rate	0.03668
household total	1.41421	children with the SUD penalty	0.03863
		children with the street penalty	0.00860
the childhood being lived		income shift, SUD penalty	0.56042
mean consumption c	0.66747	income shift, street penalty	0.69000
mean housing h	0.26338	income shift, both	0.38669
share on the street	0.01341		
share whose parent uses	0.08695	**the levels**	
$u(X_K)$	-7.54356	childhood, goods $(\xi - 1)u$	-3.12465
$u(X_P)$	-7.53432	childhood, $(\xi - 1)\aleph$	4.55813
		childhood, total	1.43349
masses at a date		adult value V_1	8.19940
children	1.00000	lifetime	9.63289
young	1.00000	childhood's share	14.88%
old	0.90672	parent books, $\omega_c(\xi - 1)u(X_K)$	-0.04743

For scale, one per cent of consumption in every state is worth 0.0613 in these units at the adult value, so the parent's own valuation of its child, -0.047, is below one per cent of consumption — which is what makes the damage close to a pure externality at the fitted ω_c.

Parameters: $\varphi_c = 0.48189$, $\varphi_h = 0.37355$, $\varphi_q = \theta = 0.14456$, $\sigma = 2$, $\aleph = 11.00431$, $\bar{b} = 0.05785$, $\omega_c = 0.01518$, $\phi_0 = 0.91$, $\kappa = 0.09839$, $\delta_d = 0.57906$, $\rho = 0.35$. Vector `calibrated_sep13.json`, iteration 1350, which carries $\beta = 1.0$.